

coil placed close to it. Rectifiers are used to convert this alternating current into a direct current to recharge the batteries.

History

Wireless charging technology has been around since the late 19th century when Nikola Tesla demonstrated magnetic resonant coupling—the ability to transmit electric energy through the air by creating a magnetic field between two circuits. The technology was first used by Hutin and Le-Blanc in 1894 to power an electric vehicle. For about hundred years the technology did not draw much attention, except perhaps for application in a few devices such as electric toothbrushes. The emphasis was on improving the performance of the internal combustion engines which were in extensive use in the vehicles, and no attention was paid towards this technology.

In 1972, Professor Don Otto of the University of Auckland revealed an electric vehicle powered wirelessly by using transmitters placed under the road and a receiver on the vehicle. In 1977, Trombly patented an electromagnetically coupled battery charger to charge headlamp batteries for miners. The real breakthrough came in 1978 when the US engineers Bolger, Kirsten, and Ng created an electric vehicle using a 180Hz wireless charging system with 20kW power.

In California, in the 1980s, inductive charging was used to power a bus, and similar work was done in France and Germany around this time. In 2006, MIT demonstrated transmission of a large amount of power over a few metres using resonant coupling. In 2007, MIT physics professor Marin Soljačić displayed wireless transfer of electricity at two metres with much improved efficiency. Soljačić established WiTricity to commercialise the technology.

The first wireless chargers appeared in the market about six years ago using tightly coupled inductive charging in which the device is required to be placed in an exact position on a pad to charge. The world's first complete hands-free plugless proximity inductive charging



Plugless Power—world's first complete hands-free proximity inductive charging system (Credit: <https://cleantechnica.com>)

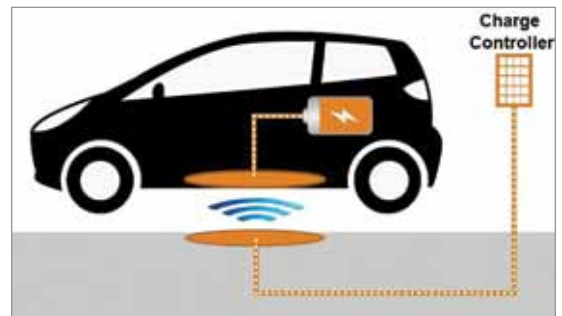
ing system for electric vehicles, Plugless Power, was brought by Google from Evatran in 2011 for their electric vehicles to transport employees. In November 2011, the Mayor of London, Boris Johnson (now the prime minister of the UK) carried out a trial on fifty EVs using thirteen wireless charging points.

In October 2014, the University of Utah added

an electric bus to its fleet that used an induction plate at the end of its route to recharge the batteries. In January 2015, Hughes Electronics developed the Magne Charge interface for General Motors and Toyota; AUDI presented a 3.6kW inductive charger during the 66th International Motor Show, and in Germany, Bombardier-Transportation presented a 3.6kW wireless charger for cars. Charging pads became commercially available in high volume in 2015.

Recently, the researchers at the Korea Advanced Institute of Science and Technology (KAIST) developed an electric transport system, called Online Electric Vehicle (OLEV), where a power source is placed underneath the road surface and power is wirelessly picked up by the test vehicles, moving on the power track in a convoy formation to avoid traffic congestion.

Today, there are nearly half a dozen wireless charging technologies available commercially, making inroads in the smartphones, automotive, healthcare, and manufacturing sectors with the promise of increased



A simple illustration of EV wireless charging at work (Credit: <https://circuitdigest.com>)

mobility. However, charging across a room as a commercial high-volume reality is still at least a year away.

How wireless charging works

Wireless electric charging for vehicles works quite like wireless smartphone charging, just that it is on a bigger scale—the distance being in centimetres instead of millimetres. Electric energy is transferred through an air gap from a magnetic coil in the charger to another magnetic coil fitted on to the car, when the car is parked such that the coils are aligned.

The wireless chargers are mainly of three types: chargers with charging pads that use tightly coupled electromagnetic inductive or non-radiative charging; charging bowls or through-surface type chargers that use loosely-coupled or radiative electromagnetic resonant charging; and uncoupled radio frequency (RF) chargers with trickle charging capability at distances of a few feet.

Basically, all the three operate on the principle of electromagnetic induction in which a magnetic coil (called